Cameron Emery

Embedded Systems & Hardware/Software Architect

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Hands-on Embedded & Systems Engineer who takes complex hardware and software bottlenecks from root-cause diagnosis and architecture to physical board bring-up and production. Currently serving as the sole software/firmware engineer for an automotive drive-by-wire accessibility manufacturer—spanning multi-MCU/FPGA hardware design, automotive bus reverse engineering (CAN-FD, LIN, SENT), Embedded Linux, mechanical CAD/3D prototyping, and AI-accelerated verification workflows.

Technical Capabilities

Hardware & Embedded: NXP S32K344, PIC32MK, dsPIC33, ARM Cortex-M (STM32/RP2040), i.MX8/i.MX93, Lattice iCE40 FPGA, Okika FPAA

Buses & Protocols: CAN, CAN-FD, LIN, SENT (SAE J2716), CXPI, BEAN, OBD-II, Automotive Ethernet PHY, UART, SPI, I2C

Bench & Manufacturing: KiCad Schematic/PCB Design, SMD Soldering & Rework, Oscilloscopes & Logic Analyzers, Fusion 360 / FreeCAD, 3D Printing & DFM (Draft Angles/Enclosures)

Software & AI Systems: Embedded C/C++, Python, Rust, TypeScript, Embedded Linux (Yocto/BitBake, Qt/QML), PyTorch (Sparse MoE, NVFP4), Automated Verification Harnesses

Professional Experience

Embedded Systems Engineer (Sole Software & Systems Engineer) — Electronic Mobility Controls | Aug 2025 – Present (Augusta, ME)

- **End-to-End Technical Ownership:** Serve as the company's sole software/firmware engineer across safety-critical automotive drive-by-wire accessibility systems (shifters, throttle/brake interfaces, servo gear controls), owning firmware sustaining, PCB design, hand-soldered board bring-up, vehicle integration, and field root-cause debugging.
- **Universal Interceptor Architecture (S32K + iCE40 FPGA + FPAA):** Architected a unified vehicle-interface module to replace separate vehicle-specific add-on boards; selected an **NXP S32K344** (for Automotive Ethernet PHY and multi-CAN-FD/LIN), a **Lattice iCE40UP5K FPGA** (sized for deterministic LUT-based bit-timing of SENT, BEAN, and CXPI protocols), and an **Okika FPAA** (for spoofing analog/resistive switches).
- **Fail-Safe Hardware & SENT Bus Integration:** Solved OEM shifter integration by adapting CAN/LIN control logic to operate over **SAE J2716 SENT** and designing a custom **short-to-OEM (NC relay)** interceptor PCB that keeps native vehicle wiring connected by default and only switches NO-to-COM when actively armed.
- **Embedded Linux, Voice HMI & Verification:** Built an Embedded Linux push-to-talk voice recognition controller using **NXP VIT** (with custom Yocto/BitBake recipes) isolated from motor-drive faults, refined production **Qt**/**QML** displays, and established a 129-test host-side C verification harness with MISRA guardrails.
- **Cross-Disciplinary Mechanical CAD & Prototyping:** Covered mechanical engineering responsibilities during team transitions—modeling custom enclosures, brackets, and injection-molding geometries (incorporating draft angles and assembly tolerances) in **Fusion 360** and **FreeCAD** and fabricating functional prototypes via in-house 3D printing.

Co-Founder & Lead Systems / Tooling Engineer — US Nylon LLC | Apr 2026 – Present

- Architected **Electrace**, a browser-based block schematic and multi-layer PCB layout tool in TypeScript with deterministic design-rule checking (DRC), power-rail propagation, and KiCad netlist export.
- Built automated `STEP`-to-enclosure CAD pipelines (**RIND**) and generative lattice G-code infill tooling (**GCodeLattice**) for additive nylon manufacturing and custom electronics enclosures.

Founder & Full-Stack Developer — Noctur.io & Independent Software Projects | Jan 2024 – Oct 2025

- Designed, built, and deployed full-stack web applications and REST/GraphQL APIs using Python, Laravel, React, TypeScript, and PostgreSQL/SQLite, managing authentication, database schema design, and CI/CD pipelines.

Selected Systems & Architecture Projects

OCB (Open Car Bus) & ucept — Automotive Bus Intelligence & Capture Reduction (Python, C, TypeScript, SQLite)

- Built a 20+ table multi-protocol automotive signal database (CAN, CAN-FD, LIN, SENT, J1939, OBD-II) and a bus-sniffing capture reduction pipeline that isolates active control bytes, rolling counters, and CRC polynomials from noisy vehicle traffic.

MCL4-COE-NCM — Consumer-GPU Sparse MoE & Cognitive Operator Architecture (Python, PyTorch, Hugging Face)

- Designed a custom language model architecture combining a Causal Encoder-Decoder MoE backbone, **single-softmax Cognitive Operator Ensemble** heads, `NVFP4` 2:4 sparsity, and CPU-offloaded AdamW optimizer states to train on 12 GB consumer GPUs; published `2.43B` and `1.77B` checkpoints on Hugging Face (`huggingface.co/mesmac`).

Kadmium (Cd-48) — Heterogeneous Distributed Compute Fabric (Rust, Lean 4)

- Applied BOINC/CERN-inspired scientific cluster scheduling concepts to heterogeneous consumer hardware in Rust, using normalized model layer-sharding and Lean 4 formal verification checks to ensure bit-identical inference across split layer cuts.